Figure 4.4 Progression pattern and post-progression cause of death

Analysis cohort: 1786 patients with first radiological progression (HAIC-Sys 733, TACE-Sys 613, Systemic 440); 1186 deaths after progression.
Deaths by cause: intrahepatic HCC 696, extrahepatic HCC 130, systemic HCC 272, non-HCC 80, undetermined 8.
Median post-progression follow-up (reverse KM): 18.7 months. Time zero = date of first progression; main time point 12 months.
Bootstrap: 1000 successful patient-level resamples (all models refitted in each). Engine check vs riskRegression: max diff 8.9e-16.

Contents:

1. Figure 4.4A adjusted 12-month CIF matrix + cause composition
2. Figure 4.4B HR / csHR matrix (reference IH-NL) + overall Wald + Lunn-McNeil
3. Cohort flow, events and numbers at risk; 4. Lunn-McNeil and follow-up tests; 5. Overall Wald tests, PH tests
6. Validation vs original IH-TG matrix; 7. Treatment interaction; 8+. Supplementary figures (sHR, ARD, PMR, unadjusted AJ, 6/24 months, Firth, sensitivity)

A Adjusted 12-month cumulative incidence of death after first progression

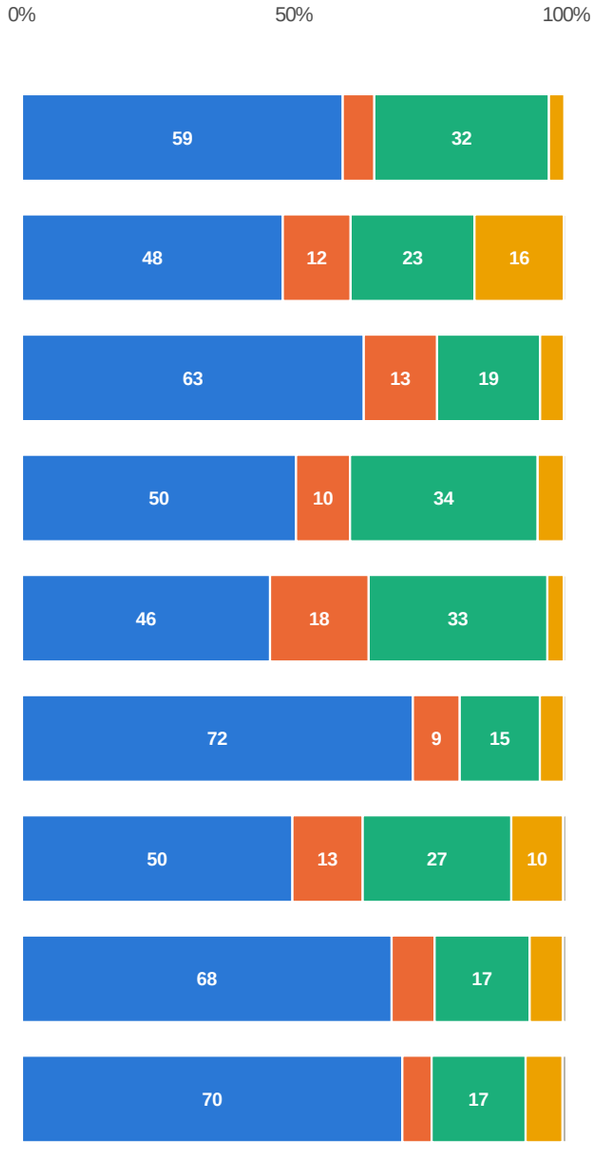
Cause-specific Cox + G-computation (standardised to all 1786 progressors); % (95% bootstrap CI, B = 1000); n = events within 12 months

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
Sys-TG	79.1 (70.5–88.1) n=55	46.6 (35.9–57.4) n=31	4.6 (0.8–11.6) n=4	25.4 (15.5–36.9) n=19	2.3 (0.0–8.0) n=1
EH-TG+NL	75.4 (63.1–87.5) n=21	36.1 (20.3–51.0) n=10	9.4 (0.0–20.5) n=2	17.2 (6.6–31.2) n=6	12.4 (0.0–29.7) n=3
EH-TG	74.9 (67.8–81.8) n=72	47.0 (37.5–55.9) n=42	10.1 (5.2–15.8) n=11	14.2 (8.2–21.0) n=15	3.3 (0.6–8.0) n=3
Sys-TG+NL	74.3 (68.2–80.9) n=135	37.4 (30.9–44.4) n=75	7.4 (3.8–11.7) n=12	25.6 (19.6–31.8) n=42	3.6 (1.0–7.0) n=6
Sys-NL	74.1 (67.4–81.4) n=63	33.7 (24.9–42.2) n=26	13.4 (7.0–22.6) n=13	24.3 (15.7–33.3) n=22	2.3 (0.0–6.2) n=2
IH-TG+NL	71.7 (64.7–78.7) n=74	51.5 (43.7–59.6) n=55	6.2 (2.0–11.1) n=6	10.5 (5.3–16.0) n=10	3.2 (0.7–6.5) n=3
EH-NL	64.9 (60.5–69.5) n=159	32.2 (27.4–37.1) n=76	8.4 (5.5–11.4) n=18	17.7 (13.5–22.5) n=48	6.2 (3.9–9.2) n=15
IH-TG	62.1 (58.6–66.1) n=269	42.1 (37.9–46.5) n=193	4.9 (3.0–7.0) n=19	10.8 (8.1–13.9) n=38	3.8 (2.2–5.6) n=16
IH-NL	57.9 (53.4–62.6) n=161	40.5 (35.3–45.8) n=109	3.1 (1.6–4.7) n=8	10.0 (7.1–13.4) n=29	3.9 (2.3–6.1) n=15

12-month CIF (%)

0 25 50 75

Cause composition at 12 months

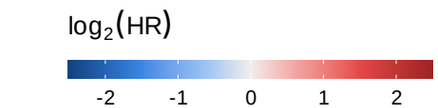


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. Rows ordered by adjusted all-cause CIF. Cause-specific CIFs plus undetermined deaths sum to all-cause. Composition labels show % of deaths by 12 months (segments <8% unlabelled).

B Post-progression death by first-progression pattern (reference: IH-NL)

All-cause: Cox HR; cause-specific: csHR (cause-specific Cox). Lunn–McNeil heterogeneity across IH/EH/Systemic HCC death: pattern × cause $\chi^2(16)$ = 60.5, P < 0.001; site × cause $\chi^2(4)$ = 48.6, P < 0.001; form × cause $\chi^2(4)$ = 7.4, P = 0.118

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.17 (0.98–1.40)	1.11 (0.89–1.38)	1.68 (0.87–3.26)	1.16 (0.74–1.82)	1.02 (0.53–1.95)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.58 (1.23–2.05)***	1.66 (1.22–2.25)**	2.53 (1.01–6.30)*	1.40 (0.73–2.69)	1.00 (0.33–3.00)
EH-TG	1.84 (1.41–2.39)***	1.58 (1.11–2.25)*	4.47 (2.14–9.36)***	2.01 (1.12–3.61)*	1.08 (0.35–3.32)
EH-NL	1.26 (1.03–1.53)*	0.84 (0.64–1.10)	3.01 (1.60–5.66)***	1.95 (1.27–3.01)**	1.68 (0.88–3.20)
EH-TG+NL	1.72 (1.10–2.68)*	1.14 (0.59–2.20)	4.09 (1.12–14.92)*	2.33 (0.99–5.48)	4.08 (1.14–14.67)*
Sys-TG	2.18 (1.62–2.95)***	1.76 (1.18–2.62)**	2.23 (0.71–6.94)	4.19 (2.38–7.38)***	NE (1 event)
Sys-NL	1.70 (1.29–2.26)***	1.03 (0.68–1.56)	5.86 (2.73–12.58)***	3.31 (1.92–5.69)***	NE (2 events)
Sys-TG+NL	1.71 (1.38–2.13)***	1.19 (0.89–1.59)	3.17 (1.53–6.57)**	3.58 (2.29–5.59)***	1.14 (0.44–2.95)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.397**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. NE: <3 events in the cell. * P<0.05, ** P<0.01, *** P<0.001 (unadjusted; BH-FDR q values in Table S-HR). Overall Wald test of the 8 pattern terms does not depend on the reference.

Cohort flow

step	n
Total patients	4204
PFS event (progression or death)	2239
excluded: death without progression	441
First radiological progression	1798
excluded: PPOS missing (PFS > OS, data error)	1
excluded: flagged for deletion (non-HCC histology / other cancer / external treatment / long LTFU)	11
Analysis cohort	1786

Deaths after first progression by pattern and cause (whole follow-up) and numbers at risk

IH/EH/Sys = intrahepatic / extrahepatic / systemic HCC-related death; Non-HCC = liver-related + other + treatment-related; Undetermined = unclassified HCC-related + unknown + missing cause

Pattern	N	IH death	EH death	Sys death	Non-HCC	Undetermined	All deaths	At risk 6 mo	At risk 12 mo	At risk 24 mo
IH-TG	476	221	24	49	21	4	319	246	101	35
IH-NL	372	134	15	34	19	1	203	234	134	58
IH-TG+NL	116	63	7	13	4	0	87	55	22	7
EH-TG	125	43	16	19	4	1	83	50	30	11
EH-NL	298	89	31	56	20	2	198	163	75	24
EH-TG+NL	34	10	3	7	3	0	23	13	7	1
Sys-TG	74	32	4	21	1	0	58	14	7	3
Sys-NL	94	28	14	23	2	0	67	34	12	6
Sys-TG+NL	197	76	16	50	6	0	148	53	30	10

Lunn-McNeil cross-cause heterogeneity tests (likelihood ratio)

Expanded data (one row per patient per HCC-related cause); non-HCC and undetermined deaths censored; strata = cause x treatment group; covariates interact with cause

test	chisq	df	p
pattern x cause (9 patterns; IH / EH / Systemic HCC death)	60.54	16	<0.001
site x cause (IH / EH / Sys site), pattern main effect kept	48.63	4	<0.001
form x cause (TG / NL / TG+NL), pattern main effect kept	7.36	4	0.118

Follow-up: pattern effect relative to weighted average pattern (weighted effect coding), per cause

Wald test (df = 2) that the effect-coded csHR is equal across IH/EH/Systemic HCC death; Holm-adjusted across 9 patterns. Descriptive counterpart: PMR (supplementary).

Pattern	IH effect-coded HR	EH effect-coded HR	Sys effect-coded HR	Wald chi2 (df=2)	P	P Holm
IH-TG	0.99 (0.88-1.12)	0.78 (0.55-1.10)	0.70 (0.56-0.89)	7.18	0.028	0.138
IH-NL	0.90 (0.77-1.04)	0.46 (0.30-0.72)	0.61 (0.45-0.81)	11.42	0.003	0.020
IH-TG+NL	1.49 (1.16-1.91)	1.17 (0.57-2.42)	0.85 (0.50-1.45)	3.61	0.164	0.534
EH-TG	1.41 (1.05-1.90)	2.07 (1.22-3.52)	1.22 (0.77-1.93)	2.33	0.312	0.624
EH-NL	0.75 (0.62-0.91)	1.40 (0.98-1.99)	1.18 (0.91-1.53)	13.25	0.001	0.011
EH-TG+NL	1.02 (0.55-1.91)	1.90 (0.58-6.15)	1.41 (0.65-3.07)	0.97	0.614	0.624
Sys-TG	1.58 (1.11-2.24)	1.03 (0.39-2.76)	2.54 (1.62-3.97)	4.03	0.134	0.534
Sys-NL	0.93 (0.64-1.33)	2.72 (1.55-4.76)	2.00 (1.32-3.05)	12.86	0.002	0.011
Sys-TG+NL	1.07 (0.86-1.33)	1.47 (0.90-2.40)	2.16 (1.63-2.88)	15.05	<0.001	0.005

Overall Wald tests of the 8 pattern terms (independent of reference group)

model	outcome	chisq	df	p
cs	All	57.89	8	<0.001
cs	IH	29.32	8	<0.001
cs	EH	30.55	8	<0.001
cs	Sys	61.93	8	<0.001
cs	NonHCC	8.38	8	0.397
fg	IH	23.41	8	0.003
fg	EH	23.20	8	0.003
fg	Sys	39.55	8	<0.001
fg	NonHCC	10.13	8	0.256
firth	All	59.72	8	<0.001
firth	IH	27.71	8	<0.001
firth	EH	32.07	8	<0.001
firth	Sys	60.17	8	<0.001
firth	NonHCC	9.40	8	0.310

Proportional hazards (Schoenfeld residuals; cs-Cox / all-cause Cox)

Where the pattern term violates PH, the HR is a follow-up-averaged effect; the fixed-time CIF/ARD (4.4A) do not rely on PH for pattern and are the primary absolute-risk summaries.

outcome	term	chisq	df	p
All	pattern	43.13	8	<0.001
All	GLOBAL	128.40	26	<0.001
IH	pattern	33.16	8	<0.001
IH	GLOBAL	100.96	26	<0.001
EH	pattern	12.17	8	0.144
EH	GLOBAL	26.56	26	0.433
Sys	pattern	9.05	8	0.338
Sys	GLOBAL	41.21	26	0.030
NonHCC	pattern	4.50	8	0.810
NonHCC	GLOBAL	29.63	26	0.283

Validation against the original IH-TG-referenced matrix

Target values from the original PDF: Sys-TG all-cause HR 1.96 (1.48-2.61), Sys-TG csHR systemic death 4.11, IH-NL all-cause HR 0.76

variant	n	deaths	allcause_SysTG	csHR_Sys_SysTG	allcause_IHNL
ORIGINAL matrix (target)	NA	NA	1.96 (1.48-2.61)	4.11	0.76
Main cohort, adjusted, strata(Group)	1786	1186	1.86 (1.40-2.48)	3.60 (2.12-6.12)	0.85 (0.71-1.02)
Main cohort, unadjusted, strata(Group)	1786	1186	1.88 (1.42-2.50)	4.66 (2.78-7.80)	0.67 (0.56-0.80)
Incl. flagged pts, adjusted	1797	1197	1.88 (1.41-2.50)	3.62 (2.13-6.16)	0.87 (0.73-1.04)
Incl. flagged pts, unadjusted	1797	1197	1.89 (1.43-2.50)	4.71 (2.81-7.88)	0.68 (0.57-0.81)
Incl. flagged, unadjusted, PPOS_status events	1797	1291	1.77 (1.34-2.34)	4.71 (2.81-7.88)	0.65 (0.55-0.77)
Main cohort, adjusted, PPOS_status events	1786	1280	1.72 (1.29-2.28)	3.60 (2.12-6.12)	0.81 (0.68-0.97)
Main cohort, baseline covariates without TTP	1786	1186	1.93 (1.45-2.57)	3.79 (2.23-6.42)	0.76 (0.63-0.90)
Incl. flagged pts, baseline covariates without TTP	1797	1197	1.94 (1.46-2.59)	3.81 (2.25-6.46)	0.77 (0.65-0.92)

Sensitivity: interaction with treatment group (justifies pooled analysis)

test	chisq	df	p
site x treatment group (all-cause death after progression)	7.18	4	0.127
pattern x treatment group (all-cause death after progression)	24.13	16	0.087

Data linkage checks

check	result
rows (raw / baseline / cod)	4204 / 4204 / 4204
raw.?? == baseline.no == cod.no	TRUE
Group identical in 3 tables	TRUE
baseline.patient_sn == cod.patient_sn	TRUE
raw.Baseline_date == baseline.????????	TRUE
deaths (Death_status_0930 = 1)	2056
deaths with non-missing Cause_of_Mortality	2040
deaths with missing Cause_of_Mortality	16
alive with a recorded Cause_of_Mortality	0
PPOS_status = 1 but Death_status_0930 = 0 (progressors)	94

Supplementary: subdistribution HR (Fine–Gray) by progression pattern (reference: IH-NL)

Fine–Gray via survival::finegray with stratum-specific censoring weights; robust SE

	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.08 (0.88–1.34)	1.50 (0.78–2.89)	1.16 (0.75–1.81)	0.98 (0.51–1.89)
IH-NL	Reference	Reference	Reference	Reference
IH-TG+NL	1.49 (1.08–2.07)*	1.82 (0.71–4.66)	1.12 (0.59–2.13)	0.80 (0.27–2.37)
EH-TG	1.10 (0.76–1.59)	3.33 (1.59–6.96)**	1.39 (0.78–2.46)	0.83 (0.26–2.66)
EH-NL	0.71 (0.54–0.93)*	2.78 (1.51–5.14)**	1.91 (1.25–2.92)**	1.62 (0.87–3.02)
EH-TG+NL	0.82 (0.42–1.59)	2.59 (0.74–9.09)	1.94 (0.84–4.48)	3.20 (0.89–11.52)
Sys-TG	1.20 (0.79–1.82)	1.40 (0.44–4.51)	2.81 (1.55–5.07)***	NE (1 event)
Sys-NL	0.74 (0.49–1.10)	4.35 (2.01–9.40)***	2.68 (1.58–4.55)***	NE (2 events)
Sys-TG+NL	0.93 (0.69–1.26)	2.14 (1.05–4.36)*	2.80 (1.77–4.44)***	0.85 (0.33–2.20)
Overall Wald P (df=8)	**P = 0.003**	**P = 0.003**	**P < 0.001**	**P = 0.256**

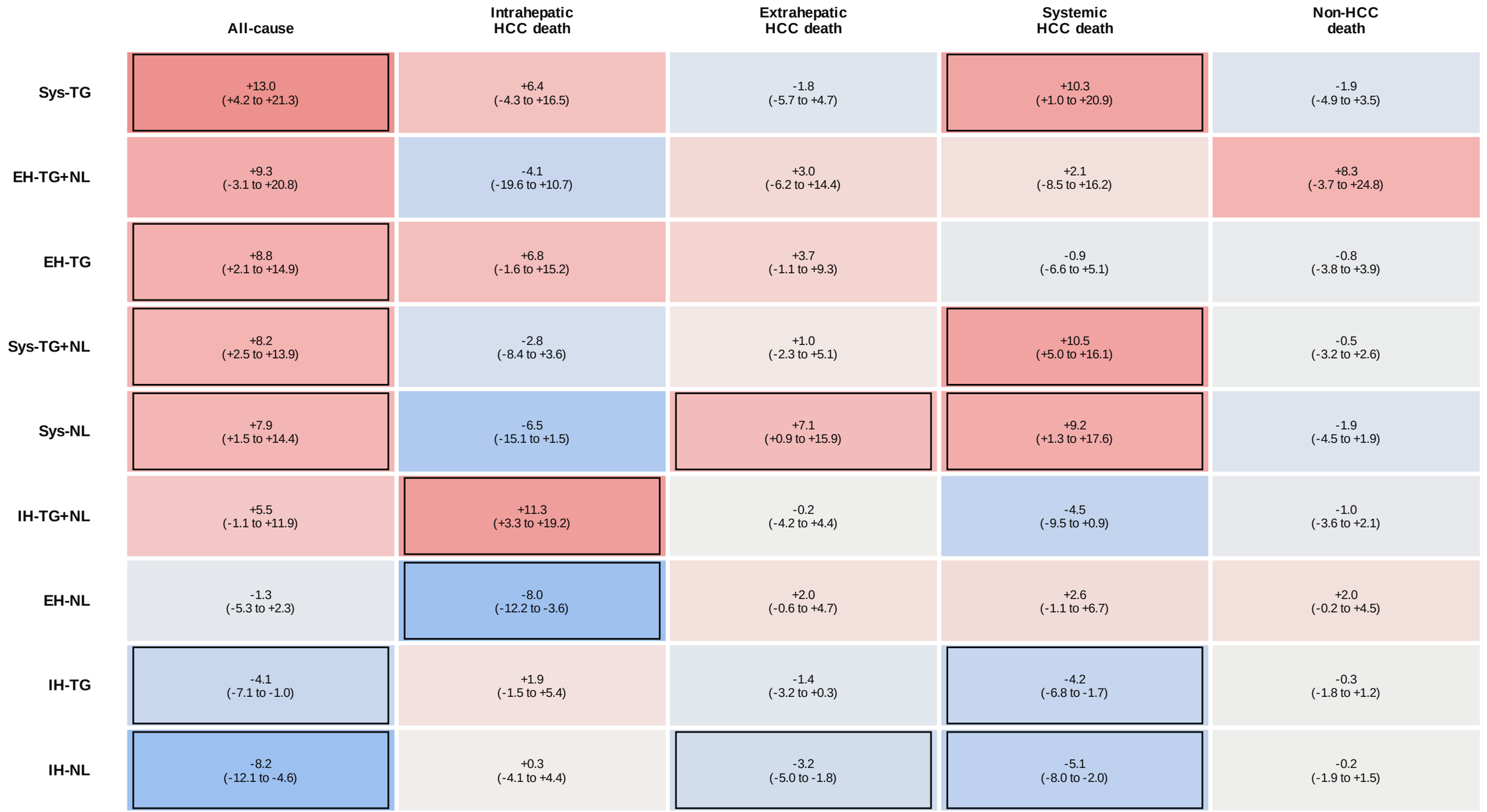
$\log_2$(sHR)

-2 -1 0 1 2

Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: absolute risk difference at 12 months vs population-standardised average

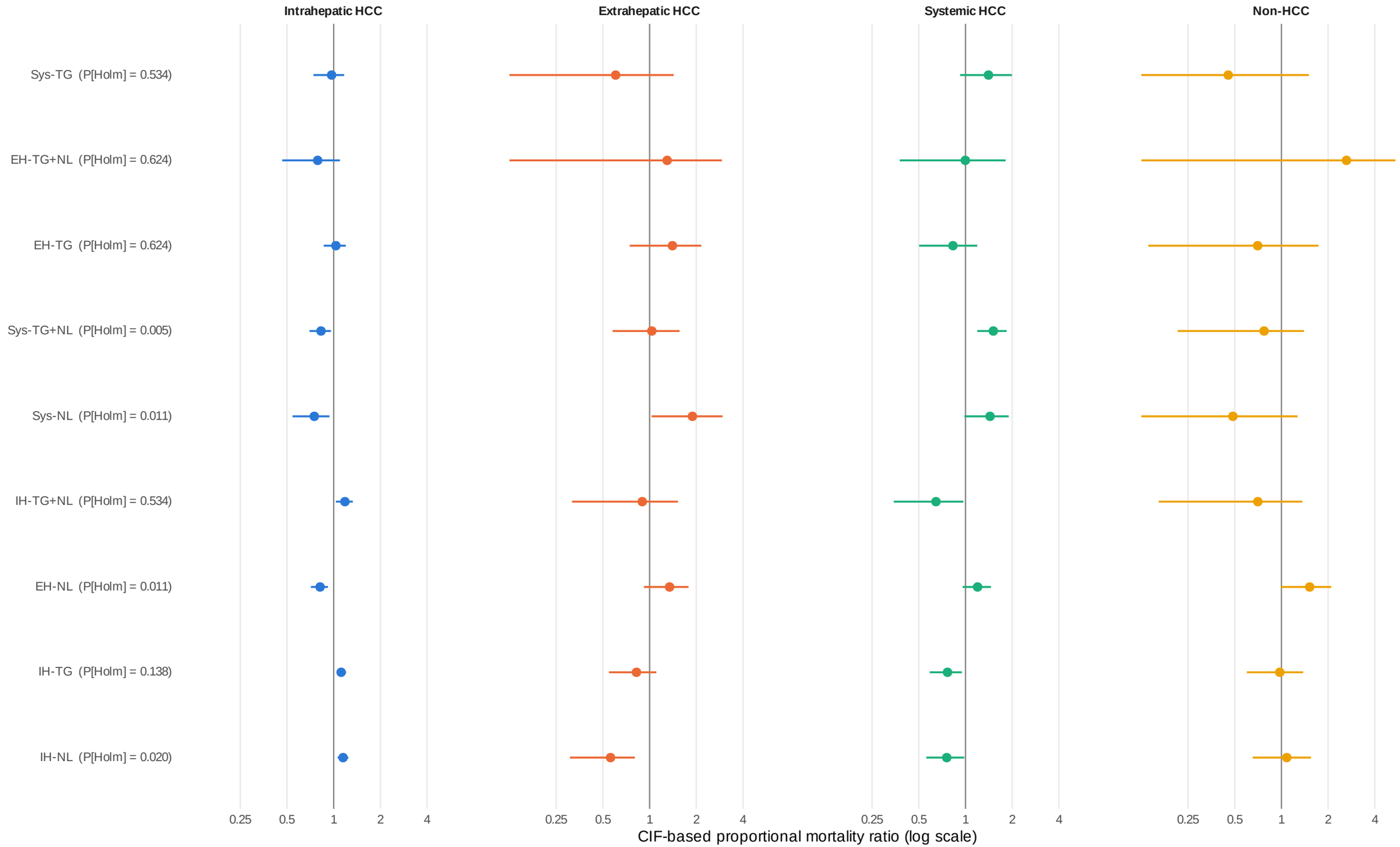
ARD = standardised CIF of the pattern − weighted average over patterns (weights = observed pattern shares); boxed: 95% CI excludes 0



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: cause-of-death enrichment (PMR) at 12 months

PMR = share of deaths from cause j within the pattern / share in all progressors. 95% bootstrap CI (clipped at 1/8 and 8).
P[Holm]: weighted-effect-coded Wald test (df = 2) that the pattern's csHR is equal across IH/EH/Systemic HCC death (from Lunn–McNeil model).

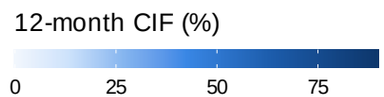


PMR is a proportion metric, not a risk metric: interpret alongside the absolute CIF matrix; PMRs within a pattern are mutually constrained.

Supplementary: unadjusted 12-month cumulative incidence (Aalen–Johansen)

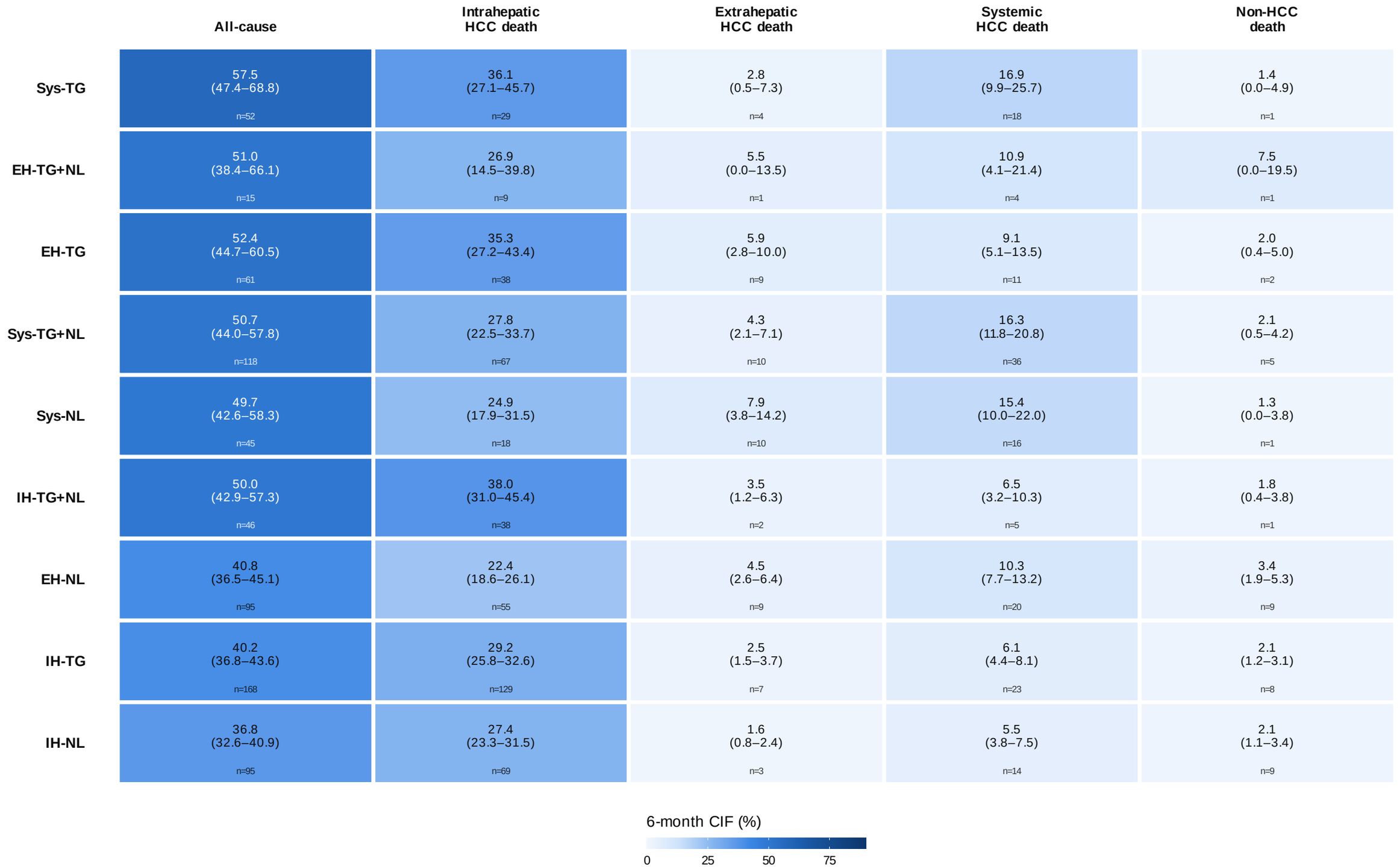
% (95% CI); all-cause = 1 − Kaplan–Meier; n = events within 12 months

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
Sys-TG	83.4 (70.5–90.7) n=55	47.5 (36.6–61.7) n=31	5.7 (2.2–14.8) n=4	28.6 (19.5–41.8) n=19	1.6 (0.2–11.2) n=1
EH-TG+NL	74.0 (51.2–86.2) n=21	35.0 (21.2–57.8) n=10	7.2 (1.9–27.3) n=2	20.9 (10.3–42.8) n=6	10.9 (3.7–31.7) n=3
EH-TG	63.4 (53.1–71.4) n=72	36.4 (28.6–46.4) n=42	10.0 (5.7–17.5) n=11	13.5 (8.4–21.6) n=15	2.7 (0.9–8.1) n=3
Sys-TG+NL	78.1 (70.7–83.7) n=135	43.3 (36.4–51.4) n=75	7.0 (4.1–12.1) n=12	24.3 (18.7–31.7) n=42	3.5 (1.6–7.7) n=6
Sys-NL	81.0 (69.2–88.2) n=63	33.0 (24.1–45.4) n=26	16.8 (10.2–27.7) n=13	28.5 (19.9–40.6) n=22	2.7 (0.7–10.6) n=2
IH-TG+NL	73.0 (62.3–80.7) n=74	53.8 (44.7–64.6) n=55	6.2 (2.9–13.6) n=6	9.9 (5.5–17.8) n=10	3.2 (1.0–9.7) n=3
EH-NL	62.9 (56.2–68.6) n=159	29.3 (24.2–35.5) n=76	7.3 (4.7–11.4) n=18	19.7 (15.3–25.5) n=48	5.9 (3.6–9.6) n=15
IH-TG	66.8 (61.7–71.3) n=269	47.4 (42.7–52.6) n=193	5.0 (3.2–7.7) n=19	9.6 (7.1–13.1) n=38	4.0 (2.5–6.5) n=16
IH-NL	49.8 (43.9–55.1) n=161	33.4 (28.6–39.0) n=109	2.6 (1.3–5.1) n=8	9.2 (6.5–13.0) n=29	4.7 (2.8–7.7) n=15



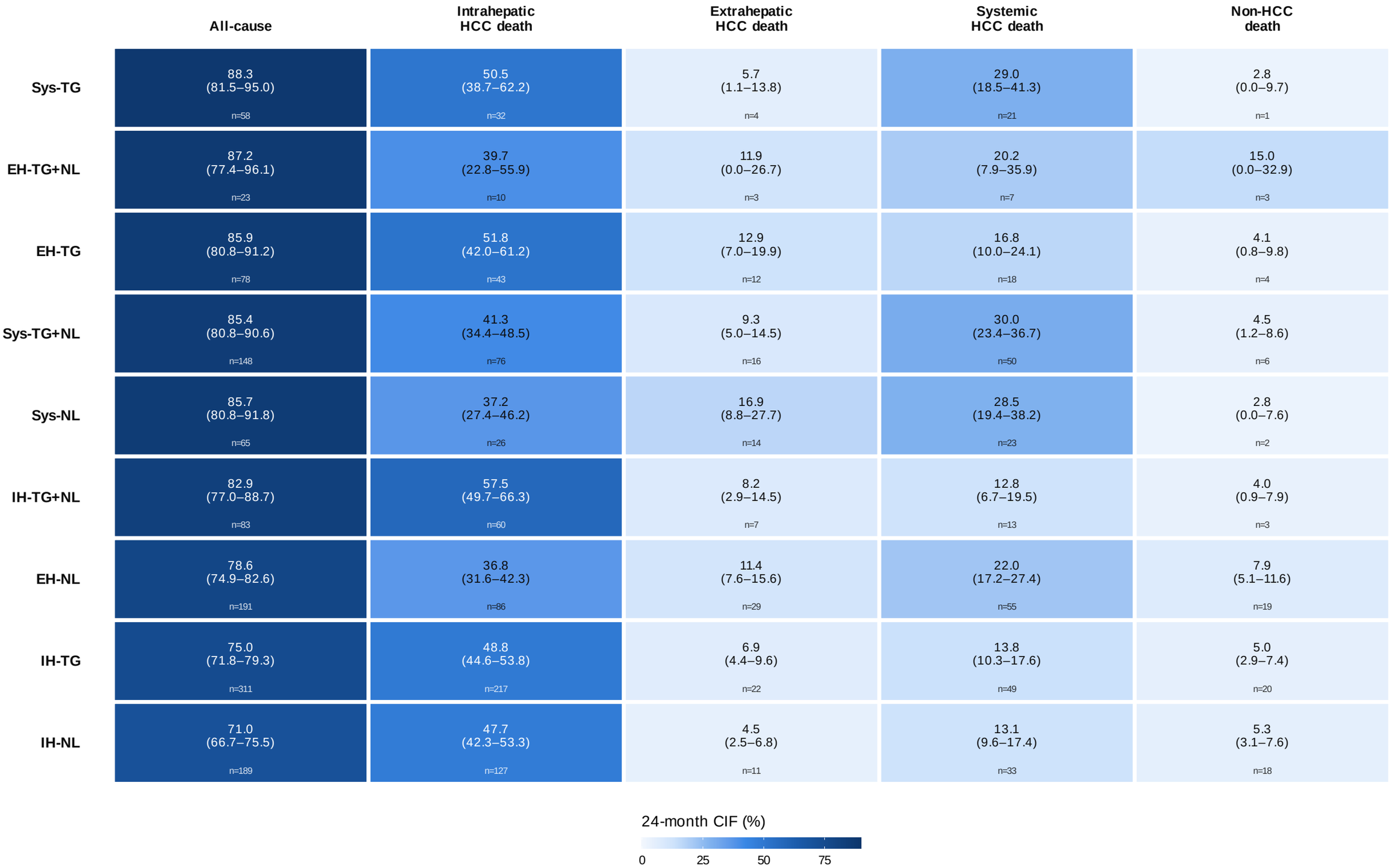
Supplementary: adjusted 6-month CIF after first progression

Same models as Figure 4.4A; n = events within 6 months



Supplementary: adjusted 24-month CIF after first progression

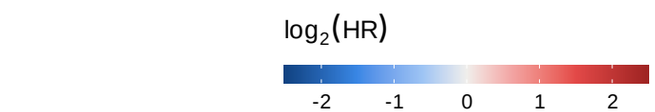
Same models as Figure 4.4A; n = events within 24 months



Supplementary: Firth-penalised HR / csHR (reference: IH-NL)

coxphf (Firth-penalised partial likelihood, Wald CI from penalised information); treatment group entered as covariate (coxphf has no strata)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.17 (0.98–1.40)	1.10 (0.89–1.37)	1.68 (0.87–3.26)	1.18 (0.75–1.84)	1.01 (0.52–1.95)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.55 (1.20–2.00)***	1.59 (1.17–2.16)**	2.58 (1.05–6.34)*	1.46 (0.76–2.77)	1.07 (0.37–3.13)
EH-TG	1.81 (1.39–2.35)***	1.55 (1.09–2.20)*	4.48 (2.14–9.36)***	2.06 (1.16–3.66)*	1.19 (0.40–3.55)
EH-NL	1.26 (1.04–1.54)*	0.84 (0.64–1.11)	3.06 (1.63–5.73)***	1.96 (1.27–3.01)**	1.64 (0.85–3.16)
EH-TG+NL	1.78 (1.15–2.76)*	1.24 (0.65–2.36)	4.84 (1.42–16.56)*	2.38 (1.04–5.47)*	5.04 (1.48–17.22)**
Sys-TG	2.22 (1.64–2.99)***	1.82 (1.23–2.70)**	2.49 (0.82–7.54)	4.10 (2.33–7.21)***	NE (1 event)
Sys-NL	1.75 (1.32–2.32)***	1.08 (0.71–1.63)	6.18 (2.90–13.18)***	3.35 (1.95–5.74)***	NE (2 events)
Sys-TG+NL	1.73 (1.40–2.15)***	1.22 (0.91–1.62)	3.09 (1.50–6.41)**	3.52 (2.25–5.49)***	1.25 (0.48–3.23)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.310**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR excluding undetermined deaths

n = 1778 (8 deaths with unclassified/unknown/missing cause removed)

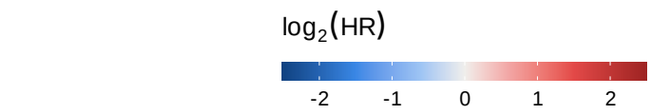


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR including patients flagged for deletion

n = 1797

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.14 (0.96–1.37)	1.09 (0.87–1.35)	1.68 (0.87–3.26)	1.16 (0.74–1.81)	1.01 (0.53–1.93)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.57 (1.21–2.02)***	1.65 (1.22–2.25)**	2.54 (1.02–6.34)*	1.43 (0.75–2.74)	1.01 (0.34–3.01)
EH-TG	1.79 (1.38–2.34)***	1.58 (1.11–2.24)*	4.52 (2.16–9.46)***	2.04 (1.14–3.65)*	1.10 (0.36–3.36)
EH-NL	1.25 (1.03–1.52)*	0.84 (0.64–1.10)	2.98 (1.59–5.60)***	1.94 (1.26–3.00)**	1.68 (0.88–3.20)
EH-TG+NL	1.69 (1.08–2.63)*	1.14 (0.59–2.20)	4.15 (1.14–15.12)*	2.37 (1.01–5.57)*	4.14 (1.15–14.88)*
Sys-TG	2.15 (1.59–2.90)***	1.75 (1.18–2.60)**	2.24 (0.72–6.98)	4.19 (2.38–7.38)***	NE (1 event)
Sys-NL	1.68 (1.27–2.22)***	1.03 (0.68–1.56)	5.89 (2.74–12.64)***	3.32 (1.93–5.71)***	NE (2 events)
Sys-TG+NL	1.67 (1.34–2.07)***	1.16 (0.87–1.54)	3.15 (1.52–6.53)**	3.49 (2.23–5.46)***	1.13 (0.44–2.94)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.385**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.